# LAB 03 — AI Data Source Inventory & Evaluation

Module 03: AI Data Landscape — Sources, Repositories & Data Houses

SLOs Assessed: 1, 4 | Estimated Time: 90 minutes | Submission Format: This completed document

## Student Information

| | | |
|---|---|---|
| **Student Name:** | Williane Yarro | |
| **Date Submitted:** | 6/22/26 | **2001/14060** |

## Lab Overview

**This lab maps directly to Module 03 SLOs:**

- SLO 1 — Classify AI data sources by type, origin, and access model
- SLO 4 — Navigate data discovery tools, metadata standards, and dataset documentation practices

You will complete four sequential activities. Each builds on the previous: choose a domain, discover datasets using real tools, evaluate them systematically with the DAQCE framework, then write a proper Datasheet for your top choice.

## Part A — Domain Selection & Problem Statement

Objective: Define exactly what AI task you need data for before you start searching.

### A1. Choose Your AI Domain

Select ONE domain from the options below, or propose your own. Circle or bold your choice in this document.

a) Healthcare NLP — clinical note classification, medical entity recognition

b) Autonomous Vehicles — object detection, pedestrian recognition, road segmentation
c) Financial Fraud Detection — transaction anomaly detection, credit risk scoring
d) Sentiment Analysis — social media monitoring, product review classification
e) Agricultural AI — crop disease detection, yield prediction from satellite imagery
f) [b ] My own domain: ____ Autonomous Vehicles — object detection, pedestrian recognition, road segmentation ____________________________________________

## A2. Problem Statement (2–3 sentences)

Write a precise, model-ready problem statement. Include: the AI task type, the data modality needed, and the intended deployment context.

This initiative seeks to create a multi-task deep learning framework for real-time object detection, pedestrian identification, and road segmentation in self-driving vehicles, utilizing RGB camera images as the main data source.

The model is required to precisely recognize vehicles, pedestrians, and navigable road sections within intricate urban , traffic management and highway settings. It will be tailored for implementation on embedded automotive hardware to facilitate safe, low-latency decision-making in Level 2+ autonomous driving systems.

## A3. Data Requirements Specification

Complete this table before searching. It will guide your evaluation in Part B.

| | |
|---|---|
| **Required Modality** | Multimodal expel: images,GPS data, video, sensor/Lidar |
| **Target Language(s)** | Metadata in English, no dependant language |
| **Label Type Needed** | Bounding boxes, object classifications such as : cars, pedestrians, traffic signs, semantic segmentation labels and lanes making |
| **Minimum Record Count** | s ≥ 500,000 labeled images/frames |
| **Date Range Required** | 2020-2025 vehicules technology and driving environment |
| **Geographic Scope** | Rural, suburban, and urban environment |

| | |
|---|---|
| **License Requirement** | e.g. Commercial and open license permitting research use CC BY, CCO, and equivalent |
| **PII Restrictions** | e.g., No personal information should be displayed. Face and license plates should be blurred and anonymized. |

## Part B — Dataset Discovery & Comparison

Objective: Use at least three different discovery tools to locate three candidate datasets. Document each systematically.

### B1. Discovery Tools Used

List every tool and query string you used during your search. Minimum: three distinct tools.

| Discovery Tool | Query / Search Terms Used | Results Found (Y/N, count) |
|---|---|---|
| Hugging Face Hub | nuScenes, KITTI, autonomous driving object detection | Y(~100) |
| Kaggle Datasets | KITTI object detection, autonomous driving, road segmentation | Y (~50+) |
| UCI ML Repository | autonomous driving, vehicle, object detection | detectionY (few, mostly related can bus / vehicle data) |
| Google Dataset Search | autonomous vehicle object detection dataset, nuScenes, KITTI | Y (~1,000+) |
| OpenML | autonomous vehicle, driving, KITTI | Y (~200+) |

| | | |
|---|---|---|
| Papers With Code | autonomous driving, object detection, semantic segmentation | Y (~200+) |
| data.world | autonomous driving dataset, KITTI, nuScenes | Y(limited relevant datasets |
| Other: | https://github.com/Armanasq/kitti-dataset-tutorial | Y(high quality and official source) |

## B2. Dataset Comparison Matrix

Complete every cell for each of your three candidate datasets. Leave no blanks — write N/A where information is not available.

| Evaluation Criterion | Dataset 1 | Dataset 2 | Dataset 3 |
|---|---|---|---|
| **Full Dataset Name** | nuScene | KITTI | Waymo Open Dataset |
| **Repository / Source URL** | https://www.nuscenes.org/ | https://www.cvlibs.net/datasets/kitti/ | https://waymo.com/open/ |
| **Task / Use Case** | 3D object recognition, tracking, semantic classification, and multi-tasking perception | 2D/3D object detection, road segmentation, stereo vision | 2D/3D object detection, tracking, panoptic segmentation |
| **Data Modality** | RGB cameras (6), LiDAR, RADAR (multimodal) | Stereo RGB cameras, LiDAR, GPS | High-res cameras (5+), multiple LiDARs |
| **Record / Sample Count** | ~1.4 million camera images, 40k keyframes | ~7,481 training images (object detection) | Hundreds of thousands of images + millions of annotations |

| | | | |
|---|---|---|---|
| **Train / Val / Test Split** | Official splits (700/150/150 scenes) | Standard 6k/1.5 /split | Official Large scale split |
| **Evaluation Criterion** | **Dataset 1** | **Dataset 2** | **Dataset 3** |
| **Label Types** | 3D bounding boxes (23 classes), semantic segmentation masks | 2D/3D bounding boxes, semantic segmentation road | 2D/3D bounding boxes, panoptic segmentation, tracking IDs |
| **License (SPDX)** | CC-BY-NC-SA-4.0 | CC-BY-NC-SA-3.0 | Waymo Dataset License Agreement (Non-Commercial) |
| **Commercial Use?** | Contact for commercial licensing | Non commercial only | No (Non-commercial only; contact for commercial) |
| **PII Concerns?** | Yes, license plates and faces are blurred and anonymous | Minimal faces often blurred | Yes, heavily anonymous |
| **File Format(s)** | JSON metadata, .jpeg images, .pcd LiDAR | .png images, .txt labels, .bin point cloud | TFRecord files |
| **Download Size** | ~400+ GB (full multimodal) | ~180 GB (full dataset) | Several TB (very large) |
| **Last Updated** | 2020 (with minor updates) | 2012 – 2015 stable | 2019 – 2024 release |
| **Known Limitations** | Large download and cpomplex setup | Old data, small scale and limited diversity | Resctrictive license, complexe formart and large |
| **Fitness Score (1–5)** | 5 | 4 | 4.5 |

## B3. Storage Architecture Fit Analysis

Based on your datasets' size, format, and use case, determine which storage architecture is most appropriate. Tick the 'Your project fit' row and add brief notes.

| Criterion | Data Warehouse | Data Lake | Lakehouse |
|---|---|---|---|
| **Schema approach** | Schema-on-write | Schema-on-read | Both (ACID + open formats) |
| **Primary format** | Proprietary / SQL | Raw files (CSV, JSON, img) | Delta Lake / Apache Iceberg |
| **Best AI use case** | Feature extraction, BI | Pre-training corpora | Unified train + serve |
| **Example platforms** | Snowflake, BigQuery | AWS S3, Azure ADLS | Databricks, Apache Spark |
| **Supports vectors?** | Limited | No | Via integration |
| **Governance** | Strong (SQL RBAC) | Manual / light | Strong (Unity Catalog) |
| **Your project fit** | [ ] Yes [ ] No Notes: not suitable for large raw images/LiDar | [ ] Yes Notes: good for raw sensor data | Yes best overall |

Recommended storage architecture for this project and justification (2–3 sentences):

nuScenes is the most suitable option (Fitness 5/5) for your multi-task needs, which include object detection, pedestrian recognition, and road segmentation.

# Part C — DAQCE Evaluation of Top-Ranked Dataset

Objective: Apply the DAQCE quality framework to systematically evaluate your best candidate dataset before selecting it for AI training.

## C1. Dataset Selected for Deep Evaluation

| | |
|---|---|
| **Dataset Name** | nuScenes (full official name: nuScenes: A Multimodal Dataset for Autonomous Driving) \|<br><br>n the repository |
| **Repository URL** | https://www.nuscenes.org/ |
| **Version / Date** | Specific version number or snapshot date used: nuScenes 1.0 (2020 release, with minor updates |
| **Why Selected** | One sentence: why this dataset over your other two candidates? nuScenes has been chosen as the leading option due to its superior ability to meet the multi-task demands of object detection, pedestrian recognition, and road segmentation. It offers extensive multimodal annotations, a large scale, and a variety of urban driving scenarios, especially when compared to KITTI, which is smaller and older, and Waymo, which presents more complex access issues. |

## C2. DAQCE Evaluation Matrix

For each dimension, cite specific evidence from the dataset's documentation. Do not write 'N/A' — if information is unavailable, state that explicitly and note it as a concern.

| DAQCE Dimension | Evidence from Your Top-Ranked Dataset | Score (1–5) |
|---|---|---|
| **Diversity** | Does it represent the full target population? Any underrepresented groups or missing demographics? Languages, geographies, edge cases covered? | The dataset encompasses Boston (USA) and Singapore (Asia), featuring a diverse range of locations including urban and residential areas, as well as varying times of day and night, and |

| | | |
|---|---|---|
| | | different weather conditions such as sun, rain, and clouds.<br><br>It comprises 23 distinct object classes, with a particular emphasis on oversampling less common classes like bicycles and pedestrians. dditionally, it facilitates the generalization of left and right-hand traffic scenarios. |
| **Accuracy** | Are labels correct and verifiable? What was the annotation method? Is inter-annotator agreement reported? Any known label errors? | Expert annotators from Scale AI provide high-quality annotations, which undergo several validation stages.<br><br>Comprehensive instructions for annotators are readily accessible.<br><br>Although public inter-annotator agreement statistics are not disclosed, numerous quality assurance rounds are conducted to guarantee a high level of reliability. |
| **Quantity** | Is the total record count sufficient for the model type you intend to train? Are the class distributions balanced? | ~1.4 million images captured by cameras, 40,000 keyframes, and over 1.4 million 3D bounding boxes distributed across 1,000 scenes (equating to 5.5 hours of driving). This dataset is adequate for training sophisticated multi-task models that encompass both object detection and segmentation. However, certain infrequent classes continue to exhibit an imbalance, even with the application of oversampling. |
| **Consistency** | Is the schema uniform across all records? Are label definitions stable across the dataset? Any contradictory annotations? | Uniform schema across all records (JSON metadata + standardized annotations). Stable label definitions (23 classes + 8 attributes) with consistent application via expert guidelines. |

| | | |
|---|---|---|
| **Ethics / Bias** | Are there known stereotypes, harmful content, or consent issues? Was a bias audit | aces and license plates are automatically blurred for privacy. No known harmful stereotypes reported. Data from public roads with consent considerations; vulnerable groups (pedestrians) well-represented. Non-commercial license |
| **DAQCE Dimension** | **Evidence from Your Top-Ranked Dataset** | **Score (1–5)** |
| | performed? Are vulnerable groups protected? | |

## C3. Licensing & Legal Summary

| | |
|---|---|
| **SPDX License Identifier** | e.g., CC-BY-4.0, CC0-1.0, MIT, Proprietary CC-BY-NC-SA-4.0 (with additional Dataset Terms) |
| **Commercial Use Permitted?** | Yes / No / Conditional — cite the specific license clause |
| **Attribution Required?** | Yes / No — what attribution format is required? No strictly non commercial only. Commercial license required: contacting nuScenes@motional.com |
| **Share-Alike Required?** | Yes (SA) / does derivative work require same license? Yes (SA) — Derivative works must be distributed under the same license |
| **PII / Sensitive Data** | Present / Absent / Partially redacted — Partially redacted — Faces and license plates are automatically blurred and anonymous |
| **Consent Documentation** | Available / Unavailable / Unclear — https://www.nuscenes.org/terms-of-use |
| **Usage Restrictions** | List any prohibited uses stated in the license or terms of service : \| Non-commercial use only. Prohibited for any commercial advantage or monetary compensation. Additional restrictions in Dataset Terms (e.g., no redistribution without permission |

### C4. Overall Fitness Verdict

| | |
|---|---|
| **DAQCE Total Score** | /25 (sum of five dimension scores) 23/25 (sum of 5 dimension |
| **Proceed with this dataset?** | [x ] Yes [ ] Yes with conditions [ ] No — explain below |
| **Conditions / Concerns** | If conditional or no: list specific issues that must be resolved first |
| **Recommended Next Steps** | What curation actions are needed before this dataset can be used? Begin by downloading the mini version for preliminary experimentation.<br>2. Utilize the official nuScenes devkit to implement data loaders. |

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## Part D — Datasheets for Datasets

Objective: Complete a structured Datasheets for Datasets documentation for your top-ranked dataset, following the Gebru et al. (2018) framework.

Adapted from: Gebru, T., Morgenstern, J., Vecchione, B., Wortman Vaughan, J., Wallach, H., Daumé III, H., & Crawford, K. (2018). Datasheets for Datasets. arXiv:1803.09010.

| | |
|---|---|
| **Dataset Name** | nuScenes (A Multimodal Dataset for Autonomous Driving |
| **Version Documented** | 1.0 (full release, March 2019 |
| **Documentation Author** | Williane Yarro |
| **Documentation Date** | June 23, 2026 |

### D1. Motivation

For what purpose was the dataset created? Was there a specific gap it was designed to fill? Who created it and for what primary intended use? The nuScenes dataset was developed by nuTonomy (Motional) to fulfill the demand for a comprehensive, multimodal dataset for autonomous driving. It aims to surpass the constraints of previous datasets such as KITTI by offering more extensive sensor data and annotations in a variety of urban settings.

The main objective is to facilitate the advancement and evaluation of perception systems utilized in self-driving cars.

Who funded the creation of the dataset? If there is an associated grant or project, provide details.

The nuScenes dataset was created and funded by nuTonomy (now part of Motional, a joint venture between Hyundai Motor Group and Aptiv

## D2. Composition

What do the instances that comprise the dataset represent? How many instances are there in total and per split (train / validation / test)?

The dataset comprises 1,000 scenes, each approximately 20 seconds in duration, recorded in Boston and Singapore. It includes around 1.4 million camera images, approximately 40,000 keyframes, and 1.4 million 3D bounding box annotations categorized into 23 object classes.

Does the dataset contain all possible instances, or is it a sample? If a sample, what is the sampling strategy and what population does it represent?

It includes official train/val/test splits (700/150/150 scenes)

Is there a label or target associated with each instance? Describe the label schema, annotation source, and known label error rate (if any).

Every instance has associated 3D bounding boxes and attributes (visibility, pose, etc.). Labels were created by professional annotators (Scale AI)

Is any information missing from individual instances? If so, why and how many records are affected?

No major missing information per instance; all keyframes have complete sensor data and annotations.

### D3. Collection Process

| How was each instance of data collected or acquired? What mechanisms or procedures were used (e.g., API, web scraping, sensors, surveys, crowd workers)? |
| --- |
| Data collection was conducted utilizing a specialized autonomous vehicle that was outfitted with six RGB cameras, one LiDAR unit, five RADAR sensors, as well as GPS and IMU sensors. |

| Over what timeframe was the data collected? Are there any temporal gaps or coverage discontinuities? |
| --- |
| Data collection took place over a span of several months during 2017 and 2018 in both Boston (USA) and Singapore, encompassing a variety of urban environments, weather conditions, and times of day. |

| Were any individuals or communities involved in the data collection? Were they informed of the collection and its purpose? |
| --- |
| Professional drivers operated the vehicle; no crowd workers were involved in collection. Participants (other road users) were not individually informed as data was collected in public spaces |

### D4. Preprocessing, Cleaning & Labeling

| Was any preprocessing, cleaning, or other transformation applied to the raw data? Describe in detail, including what was removed and why. |
| --- |
| Unprocessed sensor data, including images and point clouds, is supplied together with their processed counterparts |

| How were labels generated or assigned? What quality control process was used? Is inter-annotator agreement (IAA) reported? |
| --- |
| Labels have been generated by expert annotators at Scale AI, employing a multi-tiered quality assurance procedure To ensure privacy, faces and license plates are automatically obscured. Aside from standard sensor calibration and synchronization, no significant cleaning of the raw data has been conducted. |

| Are the raw (unprocessed) data available in addition to the preprocessed version? If so, where? |
| --- |
| The raw, unprocessed data can be accessed via the official nuScenes download. |

## D5. Uses

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|---|
| Has the dataset been used for any AI/ML tasks already? List the tasks and any published results. Provide citations or links. |
| The dataset has found extensive application in 3D object detection, semantic segmentation, multi-object tracking, and sensor fusion. Numerous cutting-edge studies (such as BEVFormer, PETR, CenterPoint) present their findings based on nuScenes |
| What other tasks could the dataset reasonably be used for, and what potential concerns or limitations apply in those contexts? |
| |
| Are there tasks, uses, or populations for which this dataset should NOT be used? State any known contraindications explicitly. |
| It is important to note that this dataset should not be utilized for facial recognition, individual re-identification, or any commercial purposes without obtaining explicit permission |

### D6. Distribution

| |
|---|
| How is the dataset distributed? Provide the access URL, API endpoint, or contact information for obtaining it. |
| Distributed through the official website: https://www.nuscenes.org/ |
| Under what license or terms of service is the dataset distributed? What are the conditions for redistribution of derivative datasets? |
| Licensed under CC-BY-NC-SA-4.0 + nuScenes specific terms. Derivative works must follow Share-Alike rules; redistribution of the full dataset is prohibited without permission. |
| Have any third parties imposed IP-based or other restrictions on the data associated with the instances? Describe any known legal encumbrances. |
| Licensed under CC-BY-NC-SA-4.0 + nuScenes specific terms. Derivative works must follow Share-Alike rules; redistribution of the full dataset is prohibited without permission. |

### D7. Maintenance

| Who is responsible for maintaining the dataset? Provide a contact name, organization, and email or GitHub link. |
|---|
| Managed by Motional (previously known as nuTonomy) |

| How will errors or biases discovered after release be reported, documented, and corrected? Is there a versioning policy? |
|---|
| There is no clearly visible public error reporting channel, but users may reach out to the team through the official website |

| Will the dataset be updated over time? If so, at what cadence, and how will users be notified of changes? |
|---|
| The dataset has not undergone significant updates since its initial release in 2020 (version 1.0). Additionally, there is no established public versioning policy for corrections made after the release<br><br>Because nuScenes offers the best balance for the multi-task requirements of object detection, pedestrian recognition, and road segmentation, I chose it as the top-ranked dataset over KITTI and Waymo Open Dataset. NuScenes provides rich multimodal data (six RGB cameras plus LiDAR), 23 item classes with semantic segmentation masks and detailed 3D bounding boxes, and great diversity across two cities, different lighting situations, and weather. It received the highest DAQCE score (23/25), doing very well in the areas of quantity, diversity, and consistency. Its higher annotation quality, scale (~1.4M photos), and official devkit make it the best choice for creating a reliable real-time perception model for autonomous vehicles, even if it has the same non-commercial license limitation as the other two contenders. |

## Part E — Reflection & Justification

Objective: Demonstrate critical thinking about your dataset choices and the data discovery process.

### E1. Selection Justification (1 paragraph)

Explain why you chose this dataset over your other two candidates. Reference specific DAQCE scores and licensing factors.

Because nuScenes offers the best balance for the multi-task requirements of object detection, pedestrian recognition, and road segmentation, I chose it as the top-ranked dataset over KITTI and Waymo Open Dataset. NuScenes provides rich multimodal data (six RGB cameras plus LiDAR), 23 item classes with semantic segmentation masks and detailed 3D bounding boxes, and great diversity across two cities, different lighting situations, and weather. It received the highest DAQCE score (23/25), doing very well in the areas of quantity, diversity, and consistency. Its higher annotation quality, scale (~1.4M photos), and official devkit make it the best choice for creating a reliable real-time perception model for autonomous vehicles, even if it has the same non-commercial license limitation as the other two contenders.

## E2. Data Gap Analysis

What data do you wish existed for this AI task that you could not find? Describe the gap and propose how it could be ethically filled (collection, synthesis, or augmentation).

One significant data gap is the underrepresentation of rural/highway scenarios and extreme weather circumstances (such as heavy snow, fog, or sandstorms), which are essential for developing highly reliable autonomous driving systems. Synthetic data generation using simulators like CARLA or NVIDIA Omniverse to create controlled extreme scenarios, along with data augmentation techniques (like weather simulation filters) and future collection of targeted real-world data in underrepresented regions, could be an ethical way to close this gap. During synthesis, care should be made to uphold privacy norms and prevent the introduction of new biases.

## E3. Discovery Process Reflection

Which discovery tool was most useful and why? What would you do differently in future data sourcing efforts?

When paired with the official nuScenes website, Papers With Code offered immediate access to benchmarks, model performance, and excellent documentation, making it the most beneficial exploration tool. Although Google Dataset Search was useful for preliminary research, it frequently produced results that were out-of-date or of poor quality. Instead of using general platforms like Kaggle or OpenML for future data sourcing efforts, I would start straight with domain-specific hubs (e.g., Papers With Code, official research lab websites) and the nuScenes/KITTI repositories to save time and quickly locate higher-quality, well-maintained datasets.

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## Grading Rubric

| Component | Criteria for Full Credit | Points |
|---|---|---|
| Part A — Domain & Requirements | Clear, model-specific problem statement; all 8 requirement fields completed with precise, actionable values | 15 |
| Part B — Discovery & Comparison | Three distinct tools used; all 15 matrix criteria completed for all 3 datasets; storage analysis with justified recommendation | 25 |
| Part C — DAQCE Evaluation | All 5 dimensions scored with cited evidence; licensing table fully completed; fitness verdict with specific next steps | 25 |
| Part D — Datasheets for Datasets | All 7 sections answered with substantive, dataset-specific responses (not generic); contact info and versioning addressed | 25 |

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|---|---|---|
| Part E — Reflection | Justification references specific scores; gap analysis proposes a feasible ethical collection/synthesis strategy | 10 |
| **Total** | | **100** |

Instructor feedback:

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